```
[root@legacy ~]# sar -u 1 10
Linux 2.6.29.6-217.2.3.fc11.i686.PAE (legacy)    08/11/2009

12:51:41 PM    CPU     %user     %nice   %system   %iowait    %steal     %idle
12:51:42 PM    all     13.93      0.00     36.82      0.00      0.00     49.25
12:51:43 PM    all     13.50      0.00     37.00      0.00      0.00     49.50
12:51:44 PM    all     13.50      0.00     37.50      0.00      0.00     49.00
12:51:45 PM    all     13.50      0.00     37.00      0.00      0.00     49.50
12:51:46 PM    all     12.56      0.00     38.19      3.02      0.00     46.23
12:51:47 PM    all     13.00      0.00     38.00      0.00      0.00     49.00
12:51:48 PM    all     13.07      0.00     37.19      0.00      0.00     49.75
12:51:49 PM    all     13.50      0.00     37.00      0.00      0.00     49.50
12:51:50 PM    all     14.43      0.00     36.32      0.00      0.00     49.25
12:51:51 PM    all     12.94      0.00     37.31      2.99      0.00     46.77
Average:       all     13.39      0.00     37.23      0.60      0.00     48.78
```

Figure 2: A sample sar output for CPU activity

```
[root@legacy ~]# sar -r 1 10
Linux 2.6.29.6-217.2.3.fc11.i686.PAE (legacy)   08/11/09

13:00:50    kbmemfree kbmemused  %memused kbbuffers  kbcached kbswpfree kbswpused  %swpused  kbswpcad
13:00:51      1108916   2247092     66.96    126380   1678400   8191992         0      0.00         0
13:00:52      1108860   2247148     66.96    126380   1678400   8191992         0      0.00         0
13:00:53      1108860   2247148     66.96    126380   1678400   8191992         0      0.00         0
13:00:54      1108860   2247148     66.96    126380   1678400   8191992         0      0.00         0
13:00:55      1108860   2247148     66.96    126388   1678400   8191992         0      0.00         0
13:00:56      1108860   2247148     66.96    126388   1678400   8191992         0      0.00         0
13:00:57      1108860   2247148     66.96    126388   1678400   8191992         0      0.00         0
13:00:58      1108860   2247148     66.96    126388   1678400   8191992         0      0.00         0
13:00:59      1108860   2247148     66.96    126388   1678400   8191992         0      0.00         0
13:01:00      1108860   2247148     66.96    126396   1678400   8191992         0      0.00         0
Average:      1108866   2247142     66.96    126386   1678400   8191992         0      0.00         0
```

Figure 3: A sample sar output for memory performance

```
[root@server1 ~]# iostat
Linux 2.6.18-92.el5xen (server1.example.com)     08/12/2009

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.19    0.08    0.20    0.52    0.01   99.00

Device:            tps   Blk_read/s   Blk_wrtn/s   Blk_read   Blk_wrtn
sda               5.96       536.53        38.50   16207799    1163020
```

Figure 4: A typical iostat output

```
[root@server1 ~]# iostat -p /dev/sda
Linux 2.6.18-92.el5xen (server1.example.com)     08/12/2009

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.19    0.08    0.20    0.52    0.01   99.00

Device:            tps   Blk_read/s   Blk_wrtn/s   Blk_read   Blk_wrtn
sda               5.98       538.30        38.55   16207751    1160692
sda8              0.01         0.02         0.00        668          0
sda7              0.04         0.05         0.01       1459        280
sda6              0.17         0.39         0.07      11743       2208
sda5              1.78        29.24         4.09     880469     123280
sda4              0.00         0.00         0.00          8          0
sda3              1.33        23.75         6.27     715122     188832
sda2              8.97       484.76        28.10   14595946     846088
sda1              0.03         0.06         0.00       1920          4
```

Figure 5: iostat output on the FTP server

to know about your system's configuration and capabilities.

sar

The sar utility is a part of the sysstat package, so make sure you have it installed on your machine. The command collects and reports system activity information like the disk's I/O transfer rates, paging activity, process-related activities, interrupts, network activity, memory and swap space utilisation, CPU utilisation, kernel activities, TTY statistics, etc. It can optionally also take two arguments—an interval in seconds between reports, and the number of times you want the report. But there are many other options that can be explored.

Figure 2 shows a sample output of my system's CPU activity collected by sar. I am using an interval of 1 second and I want the report to be produced 10 times using the command: sar -u 1 10

You can see from the figure that my system doesn't have too much of load on the CPU. One thing I want you to configure before using sar is to set the time in the 24-hour format, instead of the 12-hour format shown in Figure 2. This is because when we graphically chart all the information for our analysis, a 24-hour format makes more sense. You can configure an alias for sar as shown below:

```
# alias sar="LANG=C sar"
```

Now if you run the same command again you will get the time in the 24-hour format. Append the above alias in your ~/.bashrc file to keep it across reboots.

If you want to know about your memory performance using sar, issue the following command:

```
sar -r 1 10.
```

(Figure 3 shows the output on my computer.)

A much better option to analyse the reports is to save them in some file and later, using a graphical plotter like gnuplot, plot the statistics. I can run sar -r 1 10 > ~/mymeminfo & to collect the report in a file that we can look at later.

iostat

According to its man page, "The *iostat* command is used for monitoring the system input/output device loading by observing the time the devices are active in relation to their average transfer rates. The reports then can be used to change the system configuration to better balance the input/output load between physical disks." The first line generated by *iostat* is the average since the last boot, so it can be ignored.

The simplest way you can use this command is by simply running iostat without any arguments (Figure 4 shows the output on my computer). It will show you the boot